



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for Algorithmic Trading

Course Code: C728

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 October 2026	C728 adaptation of the private v10 reference: four topics, eight Python labs and AI prompt packs.	Tertiary Infotech

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Course Information

Course Title	AI Vibe Coding for Algorithmic Trading
Course Reference	C728
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	09:00 – 18:00 (1-hour lunch; 30 minutes tea breaks; 7.5 instructional hours)
Mode	Instructor-led demonstrations, AI pair programming and hands-on labs
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Specify, generate and validate Python trading code with an AI assistant.
- LO2: Validate price data, visualise indicators and construct causal signals.
- LO3: Backtest with execution delay and costs, and interpret return and risk.
- LO4: Validate parameters chronologically and operate a monitored paper-trading workflow.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — From prompt to validated signal

Time	Duration	Topic / Activity
09:00–09:20	20 min	Welcome and AI review briefing
09:20–10:00	40 min	Topic 1: Getting Started with AI Vibe Coding for Trading
10:00–10:45	45 min	Lab 1: Environment and AI workflow
10:45–11:00	15 min	Tea break
11:00–12:15	75 min	Lab 2: Fetch and validate price data
12:15–12:45	30 min	Topic 2: Analysing Markets with AI
12:45–13:45	60 min	Lunch
13:45–14:15	30 min	Visual demonstration and code review

14:15–15:30	75 min	Lab 3: Indicators and visual evidence
15:30–15:45	15 min	Tea break
15:45–17:00	75 min	Lab 4: Generate and debug trading signals
17:00–18:00	60 min	Learning reinforcement: compare evidence, peer review, extension and recap

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — From backtest to monitored paper bot

Time	Duration	Topic / Activity
09:00–09:20	20 min	Day 1 retrieval and checkpoint recovery
09:20–10:00	40 min	Topic 3: Building and Backtesting Strategies with AI
10:00–10:45	45 min	Lab 5: Build a cost-aware backtest
10:45–11:00	15 min	Tea break
11:00–12:15	75 min	Lab 6: Performance, risk and code review
12:15–12:45	30 min	Topic 4: Automating and Running Trading with AI
12:45–13:45	60 min	Lunch
13:45–14:15	30 min	Visual demonstration and code review
14:15–15:30	75 min	Lab 7: Optimise without leaking the future
15:30–15:45	15 min	Tea break
15:45–17:00	75 min	Lab 8: Broker API and monitored paper bot
17:00–18:00	60 min	Learning reinforcement: compare evidence, peer review, extension and recap

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Getting Started with AI Vibe Coding for Trading	—	Lab 1, Lab 2
Topic 02: Analysing Markets with AI	—	Lab 3, Lab 4
Topic 03: Building and Backtesting Strategies with AI	—	Lab 5, Lab 6
Topic 04: Automating and Running Trading with AI	—	Lab 7, Lab 8